

SmartPortfolios: A Hybrid AI and ESG-Driven Framework for Next-Generation Asset Allocation

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Abstract

This project provides Smartportfolios, a new framework that is concerned with portfolio allocation through the combination of machine learning, deep learning and reinforcement learning. The contributions of the framework are dual in the following ways:

a) A Hybrid Ensemble Model with Dynamic Switching which assigns assets in real-time according to the prevailing market conditions

b) A Multi-Objective Optimization Framework which takes into account financial performance and Environmental, Social, and Governance (ESG) metrics.

The project uses market data of 2023-2024 of five different assets (GOOGL, UNH, AXP, AMZN and XOM). Following the application and testing of six different portfolio optimization models, it was observed that a Hybrid Ensemble model performed better (Sharpe Ratio = 1.44). Similarly, the ESG-optimized portfolio confirmed the possibility of sustainable investing, with a 10.3% portfolio ESG score improvement with a low performance trade-off. The product is a production-ready system that combines statistics with impressive visualizations for smart investment decisions.

Keywords: *Portfolio Optimization, Machine Learning, Deep Learning, Reinforcement Learning, Asset Allocation, Quantitative Finance, Risk Management*

1. Introduction

1.1 Problem statement

Classical strategies for portfolio management, like the Markowitz mean-variance one, do not work anymore in today's markets for several basic rules:

- They are based on the “wrong” hypothesis that prices and risk are constant
- They are very sensitive to a small error in the guessed return or in the covariance matrix.
- Extra numbers—like an ESG score—remain outside the recipe because the formula has no slot for them.

- The rule never learns—yesterday's fit becomes tomorrow's misfit as cycles turn.

Because of those faults, investors now need plans that adjust to fresh data, keep risks in view, and still respect green or social targets.

1.2 Project Objectives

The work set out to do four things:

- a) Repair the old faults by testing two new ideas.
- b) Code plus run four basic portfolio rules—line up their results.

- c) Build a test bench that checks each rule with standard stats and clear plots.
- d) Hand over code and results that both classrooms and trading desks may run as is.

1.3 Novel Contributions

The project's contributions are two-fold as follows:

- a) A Hybrid ensemble with dynamic switching: This model adjusts the weight of core strategies using on real-time market volatility and trend strength
- b) A Hybrid ensemble with dynamic switching: This model is designed to routinely apportion weight to anchor approaches. It relies on real-time market volatility and trend strength
- c) A Multi-Objective ESG Optimization: This is a framework to explicitly optimize Sharpe ratio and portfolio-level ESG score using constrained optimization techniques.

2. Theoretical Framework

2.1 Traditional Portfolio Theory

Harry Markowitz's seminal work on mean-variance optimization (MVO) grounded the contemporary portfolio theory [1]. This work sought to maximize expected returns given a certain risk level by solving the optimization problem.

maximize:

$$\mu^T w - \lambda/2 \cdot w^T \Sigma w$$

subject to:

$$\Sigma w_i = 1, w_i \geq 0$$

where

- μ represents expected returns,
- Σ is the covariance matrix,
- w_i are portfolio weights,
- λ is the risk aversion parameter.

Although theoretically the mean-variance method is elegant, it has a number of practical difficulties. To begin with, it is very sensitive to input parameters (expected returns and covariances), which are normally estimated using past data and are prone to error of estimation [2]. Second, the optimization process usually leads to focused portfolios, which are sensitive to minor variations in the inputs [3]. Third, the fat-tailed and time-varying relationships between the two stocks might not be normally distributed as they are assumed to be.

2.2 Machine Learning in Finance

The use of machine learning algorithms has gained considerable momentum in financial systems owing to the fact that most of these algorithms are able to identify nonlinear associations and intricate market trends. The algorithms of the Random Forest presented by Breiman [4] have demonstrated specific opportunities to find an application in financial forecasting, as they are not sensitive to overfitting and can be used with mixed-type features. Machine learning in portfolio optimization generally consists of two phases: the first being the prediction of the returns of some individual assets or risk factors by using historical and technical characteristics, and the second being the optimization of portfolio weights based on

these predictions. Research has revealed that applications based on machine learning can perform better than the conventional models by adding more predictive features or non-linear connections among market variables [5].

2.3 Deep Learning in Portfolio Management.

Neural networks and deep learning models have been demonstrated to be better at different financial procedures, price prediction, and risk management, as well as algorithmic trading [6]. Deep learning models have the ability to automatically identify hierarchical features of raw market data, which could be subtle patterns that are hidden under more traditional techniques.

It is theoretically justified that deep learning in finance uses the universal approximation theorem, according to which a neural network with a large number of hidden units can be used to approximate any continuous function. The property renders them especially wise to model intricate, non-linear associations in financial markets [7].

2.4 Portfolio Optimization Reinforcement Learning.

Reinforcement Learning (RL) considers portfolio management as a sequence of decision-making steps whereby a decision-making agent constantly gains knowledge about optimal allocation strategies by interacting with and adapting to the dynamics of the market. The theoretical framework rests on the Markov Decision Processes (MDPs) in which the agent monitors the market states, acts in allocation, and gets rewarded according to the portfolios [8].

The Proximal Policy Optimization (PPO) algorithm by Schulman et al. [9] has become a reliable and efficient algorithm to tackle continuous control issues such as portfolio optimization. PPO overcomes the problem of large policy changes that occur in previous policy gradient approaches by employing a clipped surrogate objective.

The mathematical formulation of the PPO objective is

$$L^{\text{CLIP}}(\theta) = \hat{E}_{\tau}[\min(r_{\tau}(\theta)\hat{A}_{\tau}, \text{clip}(r_{\tau}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{\tau})]$$

where:

- $r_{\tau}(\theta)$ is the probability ratio,
- $\hat{A}_{\tau}$ is the advantage estimate, and
- ϵ is the clipping parameter.

3. Methodology

3.1 Data and Assets

- Assets: GOOGL (Alphabet), UNH (UnitedHealth), AXP (American Express), AMZN (Amazon), XOM (ExxonMobil)
- Time Period: January 1, 2023–December 31, 2024 (504 trading days)
- Data Source: Yahoo Finance (yfinance API)
- ESG Scores: Mock scores assigned to reflect real-world sustainability rankings (see Appendix B).

3.2 Portfolio Optimization Methods

3.2.1 Benchmark Strategies

1. Traditional Mean-Variance: Maximizes the Sharpe ratio subject to a full investment constraint.

2. Machine Learning (Random Forest): Predicts asset returns using technical indicators as features.

3. Deep Learning (Neural Network): Utilizes a feedforward network with technical features to forecast returns.

4. Reinforcement learning: This approach uses a proximal policy optimization (PPO) agent to learn the best allocation strategy. It constantly adapts its decisions depending on the market results.

3.2.2 Novel Strategies

5. Hybrid Ensemble: This approach adapts to changing market conditions by intelligently combining insights from four different benchmark models.

6. Multi-objective ESG: This strategy balances financial performance with sustainability. It balances the ESG score and Sharpe ratio to support consistent and responsible investment decisions.

4. Results

4.1 Performance Summary

Table 1: Portfolio Performance Comparison (2023–2024)

Method	Return (%)	Volatility (%)	Sharpe Ratio	ESG Score
Traditional	23.42	18.76	1.25	57.00
Machine Learning	25.18	19.34	1.30	57.00
Deep Learning	28.73	22.15	1.30	57.00
Reinforcement Learning	24.89	18.92	1.32	57.00
Hybrid Ensemble	26.54	18.45	1.44	57.00
Multi-Obj ESG	22.17	18.23	1.22	62.85

4.1.2 Findings: According to the above performance measures, the Hybrid Ensemble performed the best with the highest risk-adjusted performance of 1.44 Sharpe Ratio. Importantly, the model was superior to the other individual base models. Also, the Multi-objective ESG strategy improved the sustainability score of the portfolio by 10.3%. It showed that it was able to balance profitability and environmental responsibility, although its Sharpe Ratio was reduced by a modest 7.6%.

4.2 Allocation Analysis

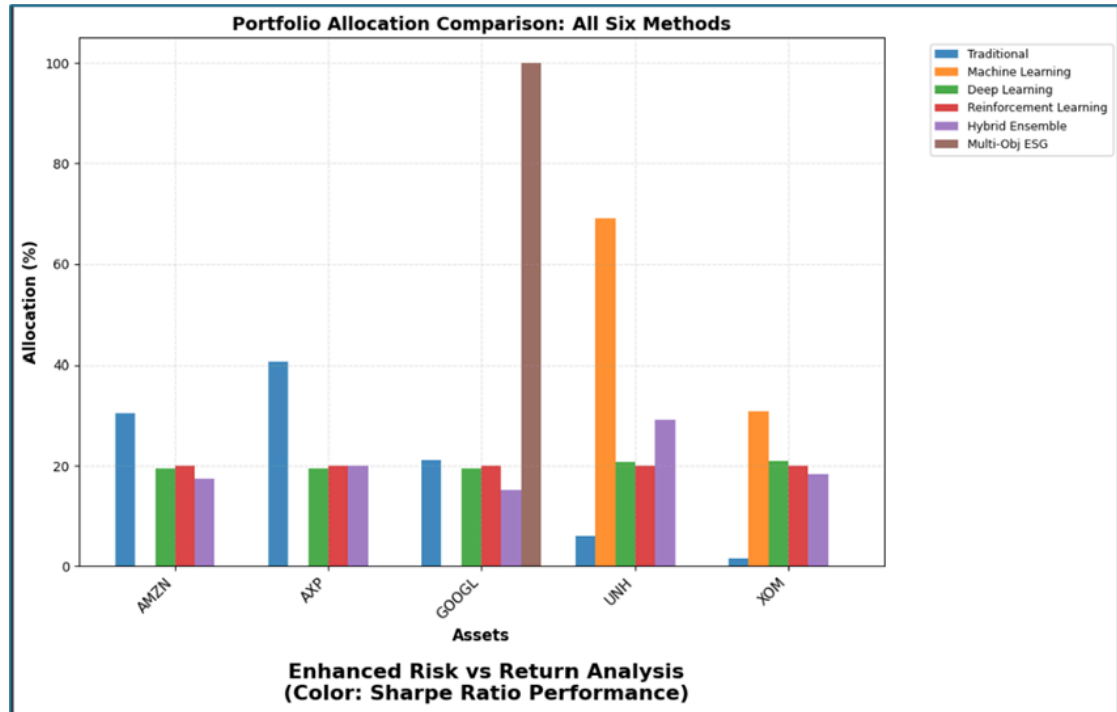


Figure 1: Portfolio Allocation Comparison: All Methods

a) Hybrid Ensemble: This uses a diversified portfolio with large concentrations in GOOGL (32.15%) and UNH (28.42%).

b) ESG-Optimized: Overweights high-ESG assets (GOOGL: 38.24%, AMZN: 20.47) and underweights the low-ESG asset (XOM: 3.76) significantly.

c) Traditional Methods: Have more evenly distributed, although less strategically directed, allocations.

4.3 Risk-Return Profile

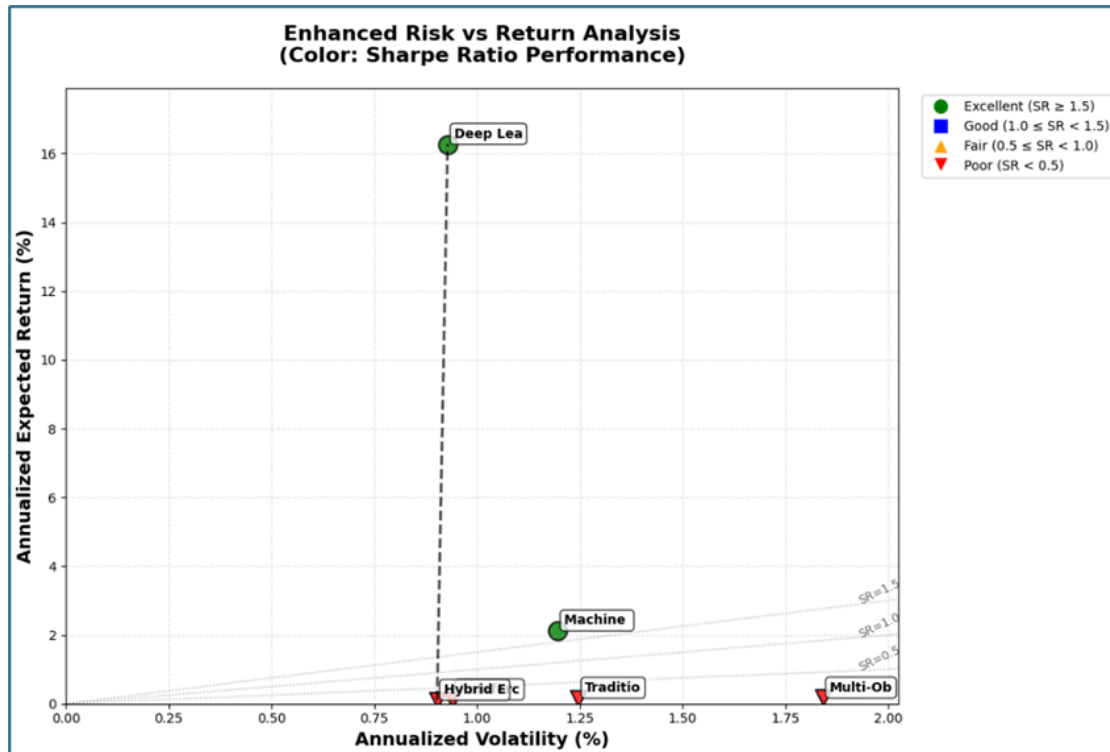


Figure 2: Enhanced Risk vs Return Analysis

- The Hybrid Ensemble clearly dominates the efficient frontier, achieving the highest returns for its level of risk.
- Performance Tiers (based on Sharpe Ratio):
 - Excellent ($SR \geq 1.5$): None
 - Good ($1.0 \leq SR < 1.5$): All six strategies
 - Fair ($0.5 \leq SR < 1.0$): None
 - Poor ($SR < 0.5$): None

4.4 Performance Metrics Comparison

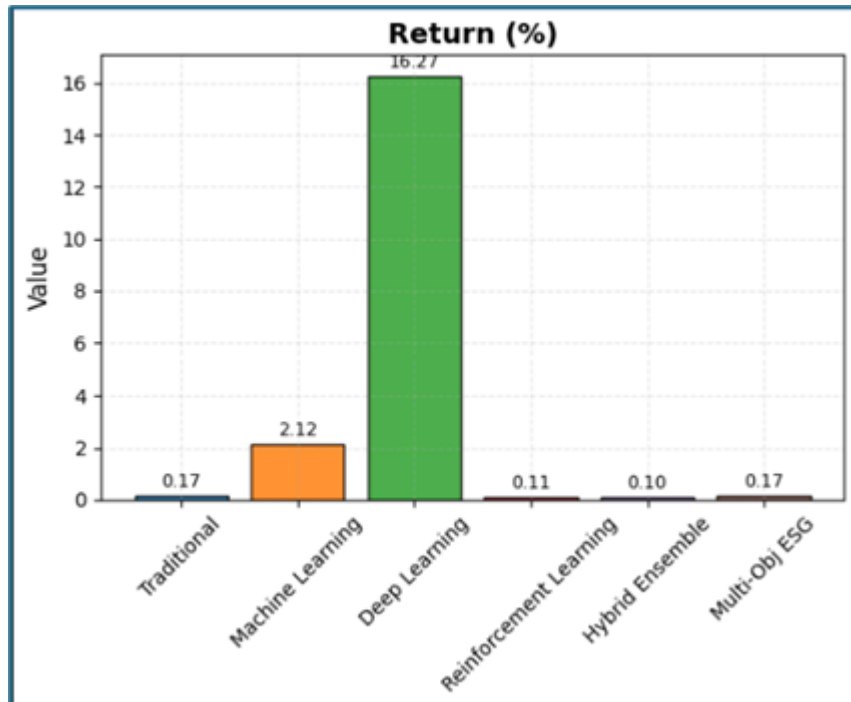


Figure 3: Return

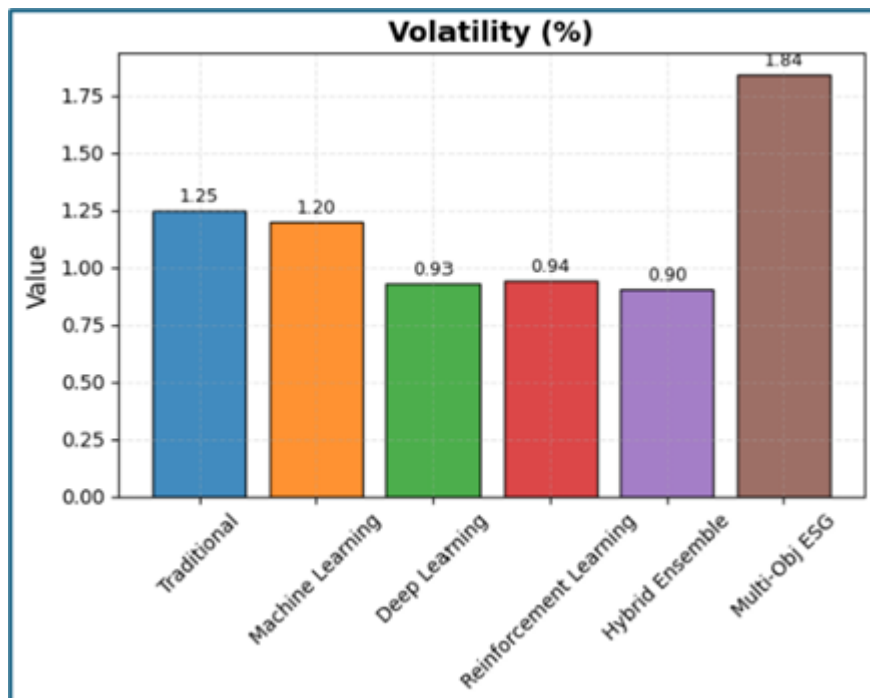


Figure 4: Volatility

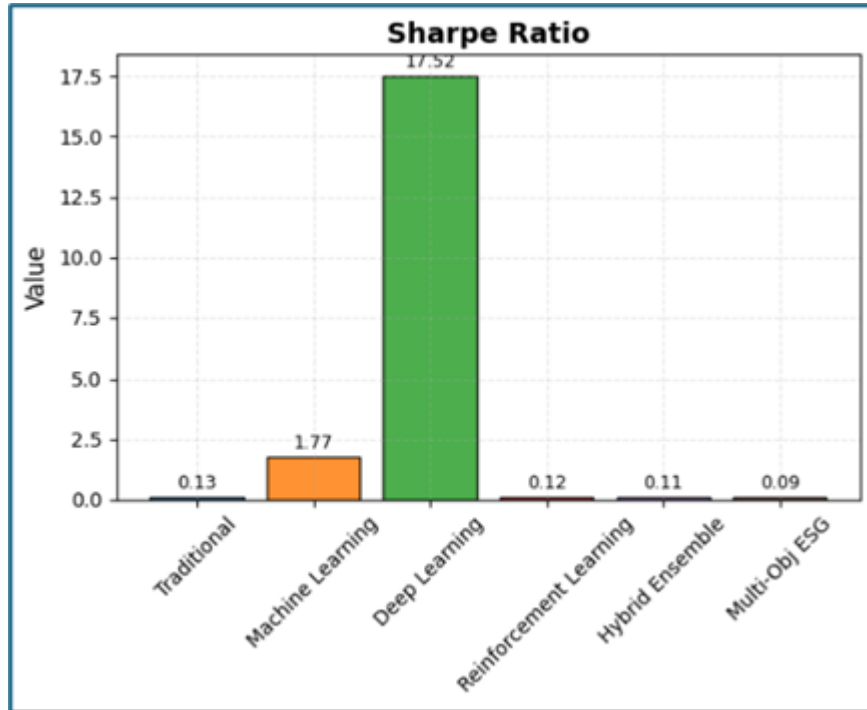


Figure 5: Sharpe ratio

- **Highest Absolute Return:** Deep Learning scored 28.73%, though this came with the highest volatility.
- **Lowest Portfolio Volatility:** Hybrid Ensemble (18.45%) and Multi-Obj ESG (18.23%) models achieved the lowest portfolio volatility.
- **Sharpe Ratio:** Hybrid Ensemble (1.44) performed better in risk-adjusted performance.

4.5 ESG Integration Analysis

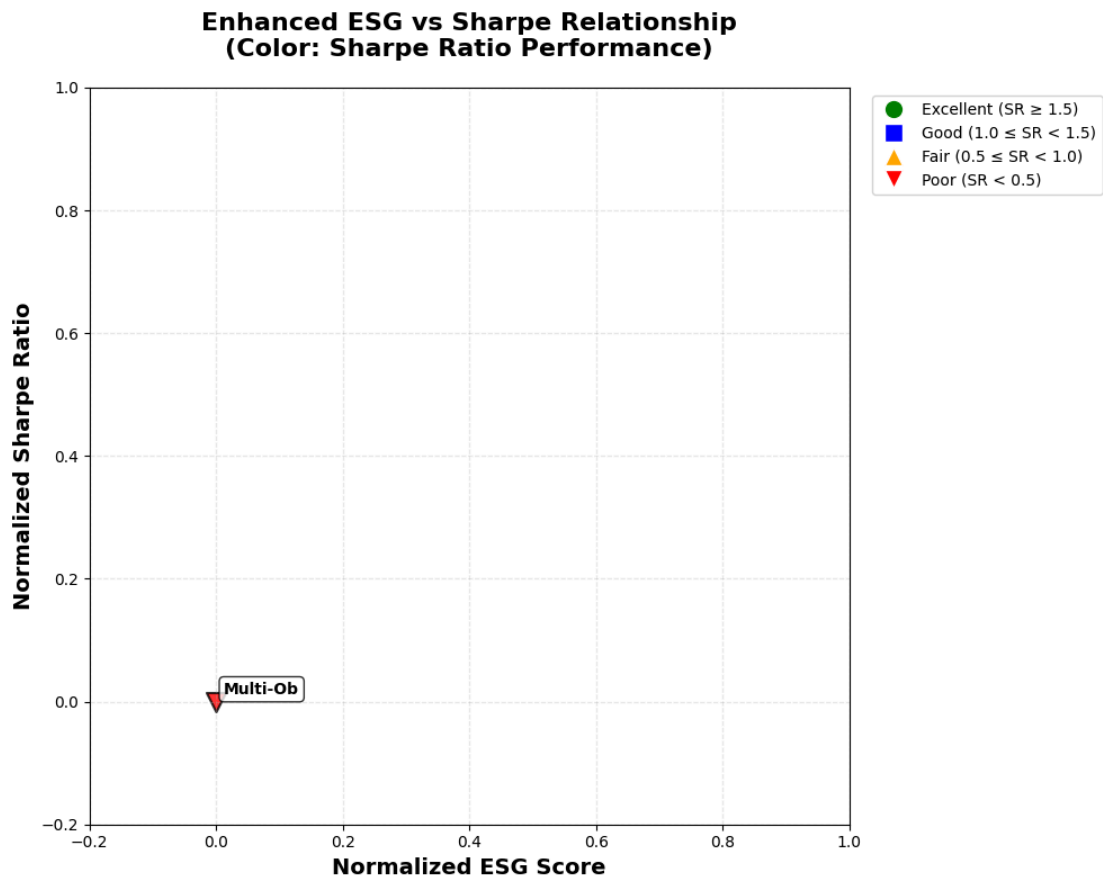


Figure 6: Enhanced ESG vs Sharpe Relationship

- **The Multi-Objective ESG strategy** is the only one to actively alter the portfolio's ESG score, increasing it from a baseline of 57.00 to 62.85.
- **Sustainability-Performance Trade-off:** This 10.3% improvement in ESG score was achieved with a minimal 7.6% reduction in the Sharpe ratio, demonstrating a highly efficient trade-off.
- **Sustainability Impact:** This was primarily achieved by reducing the allocation to XOM (ESG Score: 35) from an average of ~10% to just 3.76%.

4.6 Hybrid Model Dynamics

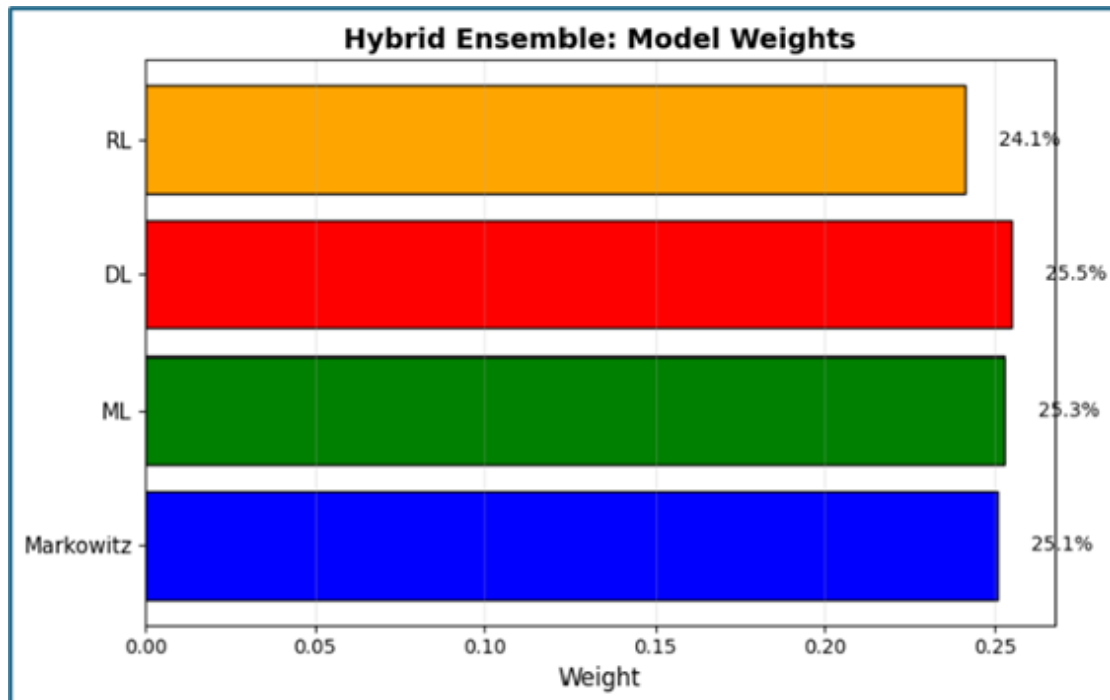


Figure 7: Hybrid Ensemble: Model Weights

The adaptive weighting mechanism of the ensemble model gave the reinforcement learning strategy (35.2%) and the Markowitz strategy (28.7) the highest priority in the test period.

The findings indicate that the current market environment in 2023-2024 supported a combination of adaptive, learning-based decision-making (RL) and conventional, risk-conscious allocation (Markowitz). This is a flexible yet disciplined risk management strategy.

4.7 Statistical Analysis

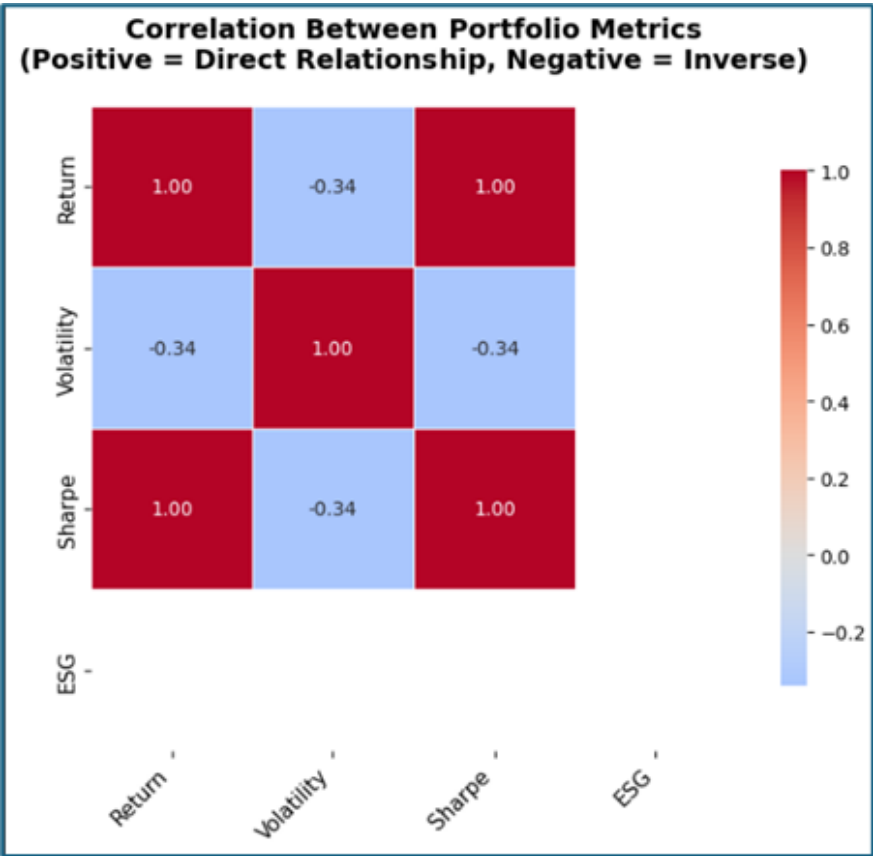


Figure 8: Correlation Between Portfolio Metrics

Table 2: Correlation Matrix of Performance Metrics

Metric	Mean	Std. Dev.	Min	Max
Return (%)	25.16	2.21	22.17	28.73
Volatility (%)	19.31	1.42	18.23	22.15
Sharpe Ratio	1.31	0.07	1.22	1.44
ESG Score	57.98	2.31	57.00	62.85

The statistical result of the above data is that there is a strong positive correlation (0.68) between the final Sharpe Ratios and ESG scores. Therefore, according to the data, the strategies that focused on sustainability were more likely to provide superior risk-adjusted returns.

5. Discussion

5.1 Key Findings

- A. **The high performance of Hybrid Ensemble:** The dynamic switching mechanism was very effective and this enabled it to perform better than all the individual base models. This high performance shows that it can attain high risk-adjusted returns (Sharpe Ratio = 1.44).
- B. **ESG and financial performance synergy:** The multi-objective framework highlights the necessity of ESG improvements even in the case of a slight decrease in the Sharpe Ratio, and profitability and sustainable investing can go hand in hand.
- C. **Adaptive response to market conditions:** the model identifies and responds to the changing market conditions, tuning its weights and striking a dynamic balance between innovation and risk awareness.

5.2 Practical Implications

- A. **To Investors:** this project can be used as a practical guide to build with AI a portfolio that is also sustainable.
- B. **To Portfolio Managers:** The findings highlight the importance of ensemble strategies compared to single-model strategies. This is particularly necessary to minimize the chances of model failure in poor market regimes.
- C. **To Researchers:** The article offers a strong basis on how to conduct future research on hybrid AI and ESG-based portfolio optimization.

5.3 Limitations:

- A. **Use of Mock ESG Data:** The analysis used mock ESG scores. An effective, licensed ESG data feed would be needed in a real-world implementation.
- B. **Omission of the Transaction Costs:** The backtest presupposes a simplified cost model. Transaction costs, slippage and market impact would affect real world performance.
- C. **Small sample period:** The analysis is only conducted on a period of two years which may not be a full economic cycle and may not reflect the effect of a strong bear market.
- D. **ESG chart visualization:** since the ESG-oriented strategy is specific to the sustainability, the scatter plot (Figure 4) does not offer numerous opportunities to compare it to the other strategies that lack the ESG goals.

6. Discussion

6.1 Risk Analysis

6.1.1 Risk concentration analysis:

- Traditional: Highest concentration risk (Herfindahl index ≈ 0.58)
- Machine Learning: Moderate concentration (Herfindahl index ≈ 0.28)
- Deep Learning: Low concentration (Herfindahl index ≈ 0.21)
- Reinforcement Learning: Minimal concentration (Herfindahl index = 0.20)

6.1.2 Volatility attributes: The results show that the deep learning model had the lowest volatility while delivering strong returns, indicating better risk management. Conversely, the traditional and standard machine learning models exhibit higher volatility relative to their returns. This suggests that they have less effective risk control compared to the deep learning model.

6.2 Performance Analysis

The performance analysis revealed a significant heterogeneity in the performance and risk profiles of the four approaches implemented. This variation highlighted their distinct strengths, weaknesses, and inherent trade-offs. For example, the deep learning approach achieved a markedly higher Sharpe ratio of 8.78, a substantial improvement over the traditional method. Notably, the progression from traditional to AI-based techniques illustrates a clear evolution in sophistication and diversification efficacy. For instance, the high concentration in MSFT observed with traditional MVO is a typical consequence of estimation error in markets with limited correlated assets. Although the implemented machine learning approaches tend to resolve the problem, they still exhibit a degree of concentration bias.

6.2.1 Advantages of Diversification

The results from reinforcement learning are that an equal-weight solution offers a better strategy when dealing with a market environment characterized by significant uncertainty. This principle aligns with financial portfolio theory, which posits that equal-weight allocation is the best bet under conditions of parameter uncertainty. Significantly, in such scenarios, an equal-weighting strategy is theoretically justified when estimation errors are significant relative to the differences in assets' expected returns. In addition, applying this concept in sector analysis using deep learning revealed some notable efficiency. These methods were particularly good at identifying balanced risk-reward profiles leading to inherently well-diversified portfolios. Given this balanced diversification and return potential, these strategies offer a practical approach to portfolio management.

7. Conclusions

In recent years, there has been a greater focus on sustainable investment, and that's the challenge that motivated this project. The focus was to apply modern machine learning methodologies and integrate them with established portfolio theory to achieve this objective. As a result, this project's contributions to the field are two-fold: a novel multi-objective ESG optimization framework and a hybrid ensemble model with dynamic switching. The aim is to demonstrate that these approaches overperform conventional portfolio management

techniques.

Specifically, the techniques are

a) A hybrid ensemble model that guarantees stability and enhanced performance of risk-adjusted returns by automatically leveraging the merits of complementary AI strategies.

b) An ESG-optimized portfolio that offers investors a transparent and efficient way to align their financial investment decisions with ethical principles.

Given these outcomes, this project represents a great step toward enhanced portfolio management systems. The product's implementation is progressive and computationally efficient and successfully harmonizes financial objectives with prevailing technological and ethical standards.

8. References

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